

Technische Hochschule Ingolstadt

Fakultät Elektro- und Informationstechnik

Sensor Network Technologies and Sensor Fusion

Optimization Approaches for Multi Object Tracking

Dec 03, 2025

Welcome to the lecture!

Prof. Dr.-Ing. Björn Giesler

Technische Hochschule Ingolstadt

AIMotion, AI and Sensor Data Fusion

Email: bjoern.giesler@thi.de

- **Please register for the lecture on Moodle!**
  - Link: https://moodle.thi.de/course/view.php?id=10580
  - Key: **SensNT+DF#WS25**
- **News on Moodle:**
  - Slides for the lecture (upcoming lectures marked as WIP and are subject to change!)
  - Formulary, Exercises (Solutions provided after discussion in the lecture)
  - Exercises are optional, one task / sheet per week, but are highly recommended.
    They can be handed in for correction.
  - I will *try* to publish exercises before Wednesday‘s lecture so that initial questions can be discussed on Thursday.
- **Exam: Written exam, 90 minutes**
- **Consultation hours: Please arrange via e-mail to bjoern.giesler@thi.de.**
- **Please also don‘t hesitate to send questions, lecture feedback, change requests, etc.!**

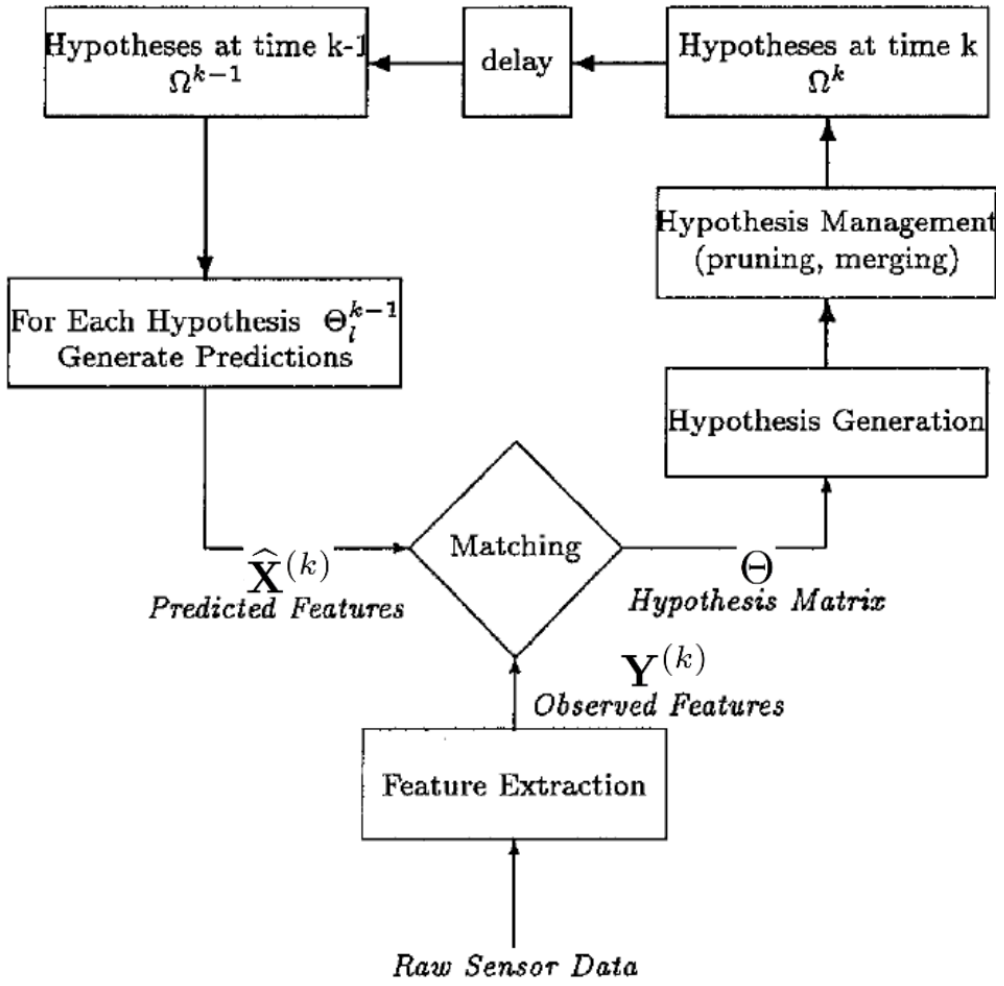
- **Like every semester, we are conducting a round of lecture / teaching feedback.**
- **Link is in Moodle, but you can also use the QR code to the right.**
- **Link gets you to the German version, but you can choose the English version (link @top right)**
- **I highly encourage feedback**
  - direct if possible (because it allows asking for clarification)
  - but this is cool too.
- **Please do participate, I'm happy to read your opinion.**



- **In the last lectures we looked at various forms of the Kalman filter – linear, extended, unscented, and explored what to do if we can't *model* a state transition function.**
- **Now we are in the middle of multi-object tracking applications.**
- **Last lecture: Multi Hypothesis Tracking.**
- **This lecture: Network Flow Optimization and other global techniques**
- **Next lecture: DNNs for tracking**
- **Going forward: Particle Filters and similar inherent multi hypothesis approaches.**
- **After that, we will revisit tracking briefly using Joint Probability Data Association Filters.**
- **The rest of the lecture series will deal with**
  - Simultaneous Localization and Mapping, and
  - (DNN-based) early fusion approaches.

Recap: Multi Hypothesis Tracking

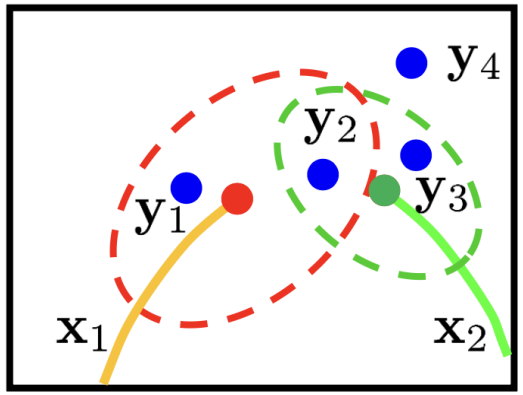
- **Keep multiple hypotheses of track assignments.**
- **Associate sequences of measurements to tracks.**
- **Evaluate the probabilities of all association hypotheses.**
- **Track each hypothesis with a standard Kalman Filter.**



Hypothesis Matrix

- **Visualize feasible associations by a hypothesis matrix**

$$\Theta = \begin{array}{c} \begin{matrix} \mathbf{x}_1 & \mathbf{x}_2 & \mathbf{x}_{\text{fa}} & \mathbf{x}_{\text{nt}} \end{matrix} \\ \begin{bmatrix} 1 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix} \end{array} \begin{matrix} \mathbf{y}_1 \\ \mathbf{y}_2 \\ \mathbf{y}_3 \\ \mathbf{y}_4 \end{matrix}$$



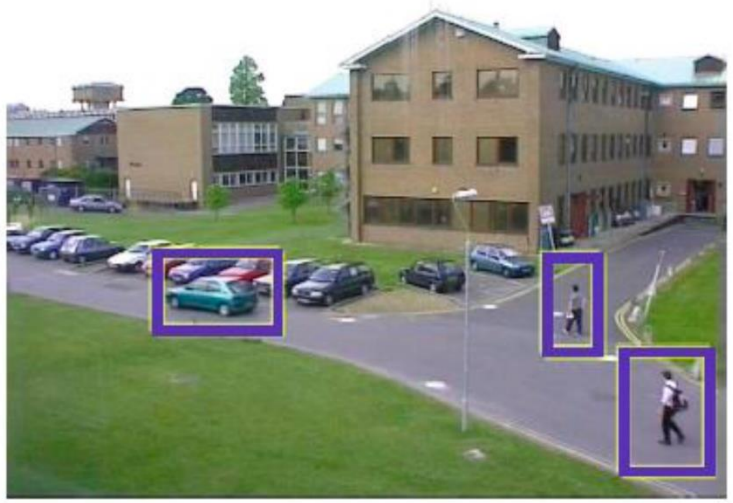
- **Interpretation**
  - Columns represent tracked objects
  - Rows represent measurements
  - A non-zero element at (i, j) denotes that a measurement $\boldsymbol{y}_i$ is contained in the validation region of track $\boldsymbol{x}_j$.
  - Extra columns $\boldsymbol{x}_{\text{fa}}$ and $\boldsymbol{x}_{\text{nt}}$ for associations as *false alarm* and *new track*.

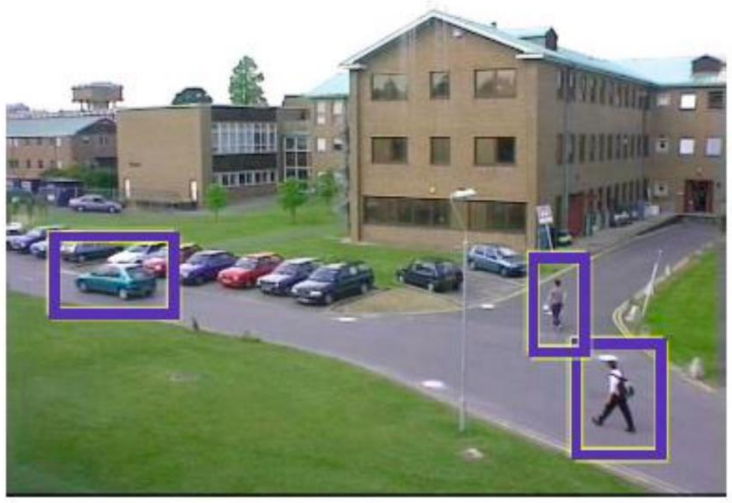
Data Association as Linear Assignment Problem

Slides credit: Robert Collins / Bastian Leibe

Back to Data Association.

Goal: Match detections across frames.









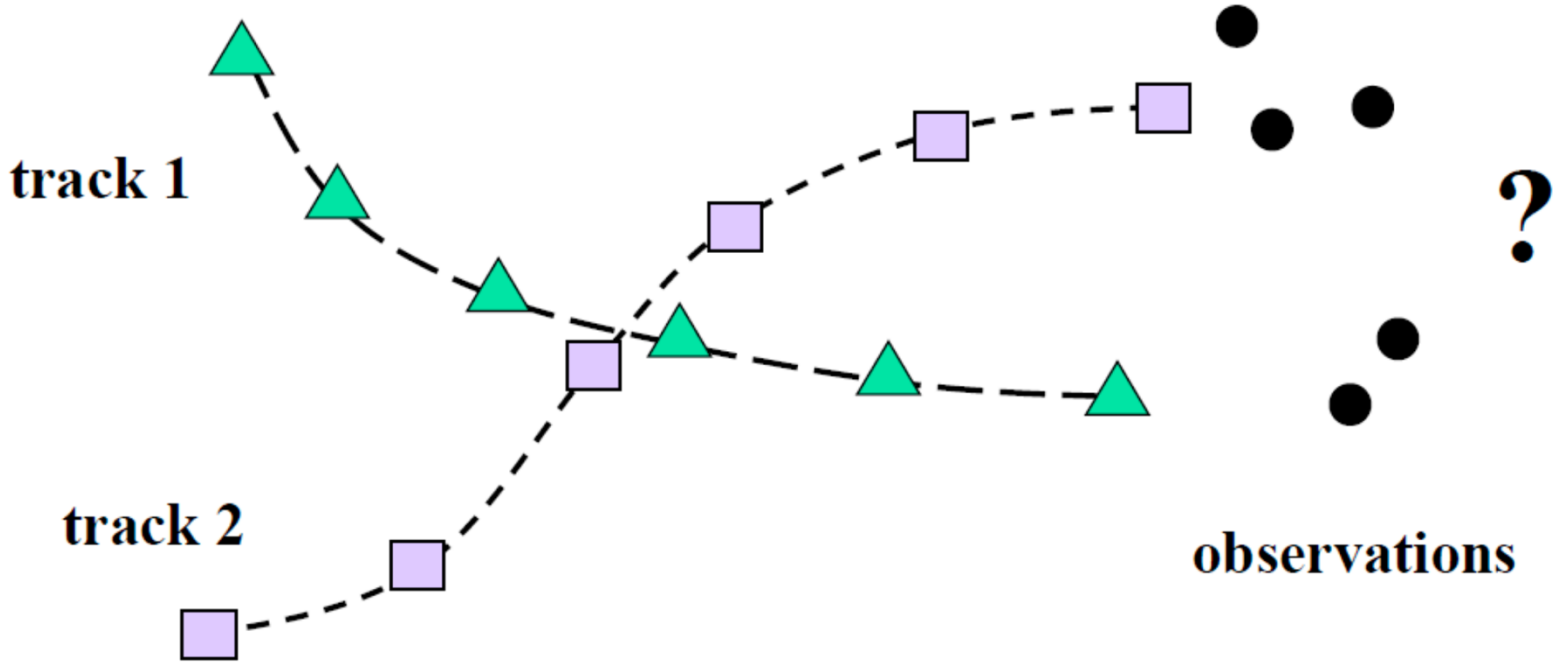








Data Association



- **Main question here:**
  - How to determine which measurements to add to which track?
  - Today: Consider this as a matching problem.

Linear Assignment Formulation

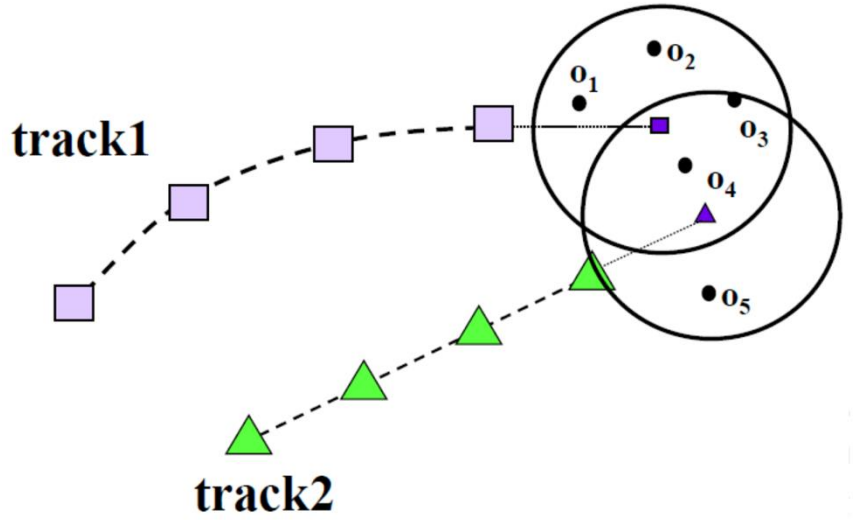
- **Form a matrix of pairwise similarity scores.**
- **Similarity could be**
  - based on motion prediction (e.g. Mahalanobis Distance)
  - based on appearance
  - combination
- **Goal:**
  - Choose one match from each row and column to maximize the sum of scores

Frame $t+1$

Frame t			
	0.11	**0.95**	0.23
	0.85	0.25	**0.89**
	**0.90**	0.12	0.81

Example: Similarity based on motion prediction

- **Predict motion for each trajectory and assign scores for each measurement based on inverse (Mahalanobis) distance, such that closer measurements get higher scores.**



	ai1	ai2
1	3.0	
2	5.0	
3	6.0	1.0
4	9.0	8.0
5		3.0

- **Choose at most one match in each row / column to maximize the sum of scores.**

Formal Definition

- **Maximize** $\sum_{i=1}^{N}\sum_{j=1}^{M} w_{ij} z_{ij}$

- **subject to**

$$\left.\begin{aligned} &\textstyle\sum_{j=1} z_{ij} = 1; i = 1, 2, \ldots, N \\ &\textstyle\sum_{i=1} z_{ij} = 1; j = 1, 2, \ldots, M \\ &z_{ij} \in \{0, 1\} \end{aligned}\right\}$$

These constraints ensure that Z is a permutation matrix.

- **The permutation matrix constraint ensures that we can only match up one object from each row and column.**
- **Note: Alternatively, we can minimize cost rather than maximizing weights --**

$$\arg\min_{z_{ij}} \sum_{i=1}^{N}\sum_{j=1}^{M} c_{ij} z_{ij}$$

Greedy Solution to Linear Assignment Problem

	1	2	3	4	5
1	0.95	0.76	0.62	0.41	0.06
2	0.23	0.46	0.79	0.94	0.35
3	0.61	0.02	0.92	0.92	0.81
4	0.49	0.82	0.74	0.41	0.01
5	0.89	0.44	0.18	0.89	0.14

- **Greedy Algorithm**
  - Find the largest score
  - Remove scores in same row and column from consideration
  - Repeat
- **Result: Sum of found scores**

Greedy Solution to Linear Assignment Problem

	1	2	3	4	5
1	0.95	0.76	0.62	0.41	0.06
2	0.23	0.46	0.79	0.94	0.35
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4	0.49	0.82	0.74	0.41	0.01
5	0.89	0.44	0.18	0.89	0.14

- **Greedy Algorithm**
  - Find the largest score
  - Remove scores in same row and column from consideration
  - Repeat
- **Result:**

score = 0.95 + 0.94 + 0.92 + 0.82 + 0.14 = 3.77

Is this the best we can do?

Greedy Solution to Linear Assignment Problem

	1	2	3	4	5
1	0.95	0.76	0.62	0.41	0.06
2	0.23	0.46	0.79	0.94	0.35
3	0.61	0.02	0.92	0.92	0.81
4	0.49	0.82	0.74	0.41	0.01
5	0.89	0.44	0.18	0.89	0.14

Greedy Score: 3.77

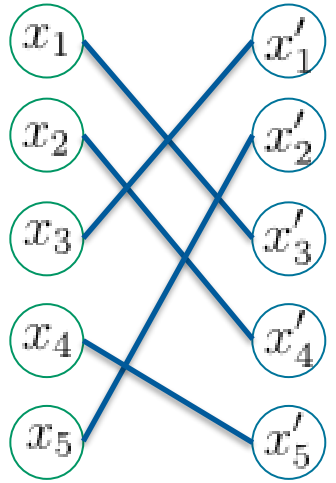
	1	2	3	4	5
1	0.95	0.76	0.62	0.41	0.06
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5	0.89	0.44	0.18	0.89	0.14

Optimal Score: 4.26

- **Greedy method is easy to program, quick to run, and yields "pretty good" results in practice.**
- **But often doesn't yield the optimal solution.**

Hungarian Algorithm

- **Combinatorial optimization algorithm to solve the assignment problem.**
- **Published in 1955 by Harold Kuhn, called "Hungarian Method" because it was based on the work of Dénes König and Jenö Egerváry – but previously published by CG Jacobi in the 1900s.**
- **Reduces the assignment problem to bipartite graph matching.**
- **Runs in $\mathcal{O}(N^3)$ for a matrix of size $N \times N$.**
- **The formulation in terms of bipartite graphs is long-ish, but for assignment matrices, it is the equivalent of**
  - permuting the columns of the assignment matrix
  - until the trace of the matrix becomes optimal.
- **→ Blackboard**

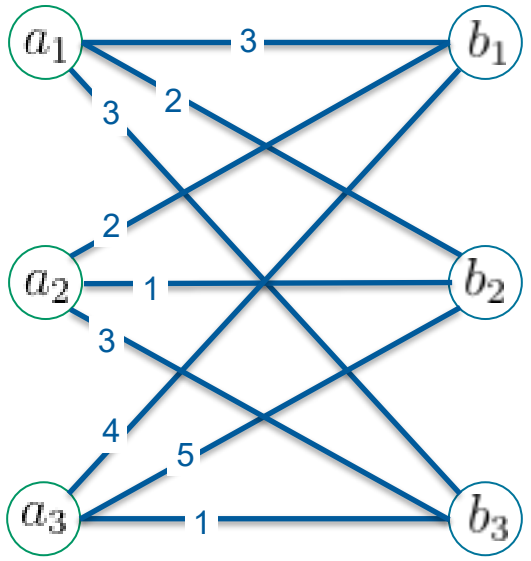


Bipartite Graph Matching

Tracking as Network Flow Optimization

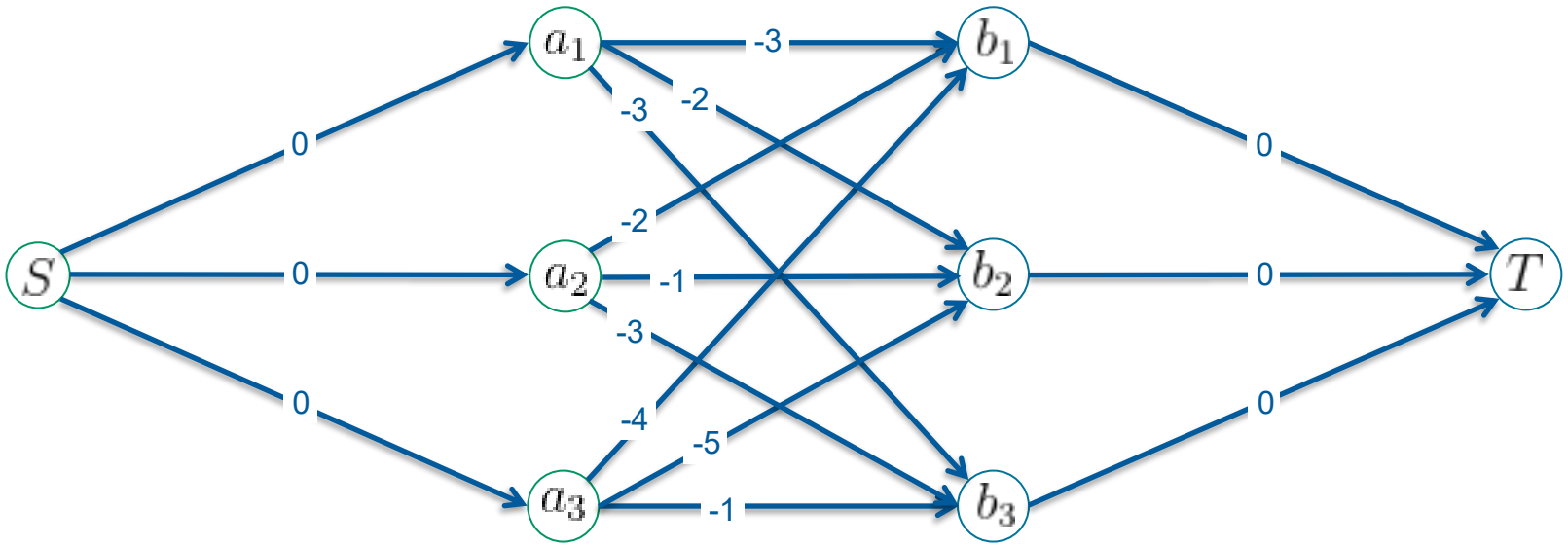
Minimum-Cost Flow – Small Example

	1	2	3
1	3	2	3
2	2	1	3
3	4	5	1



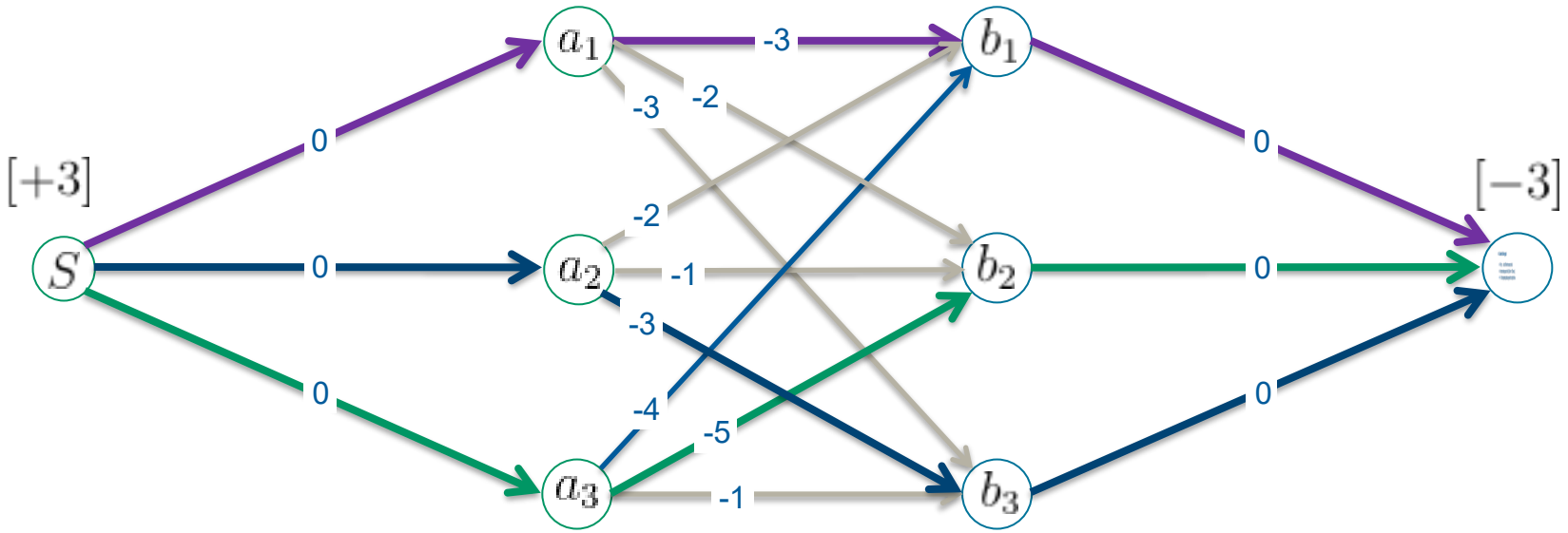
- **Network Flow Formulation: Reformulate linear cost assignment into a minimum-cost flow problem**

Minimum-Cost Flow – Small Example



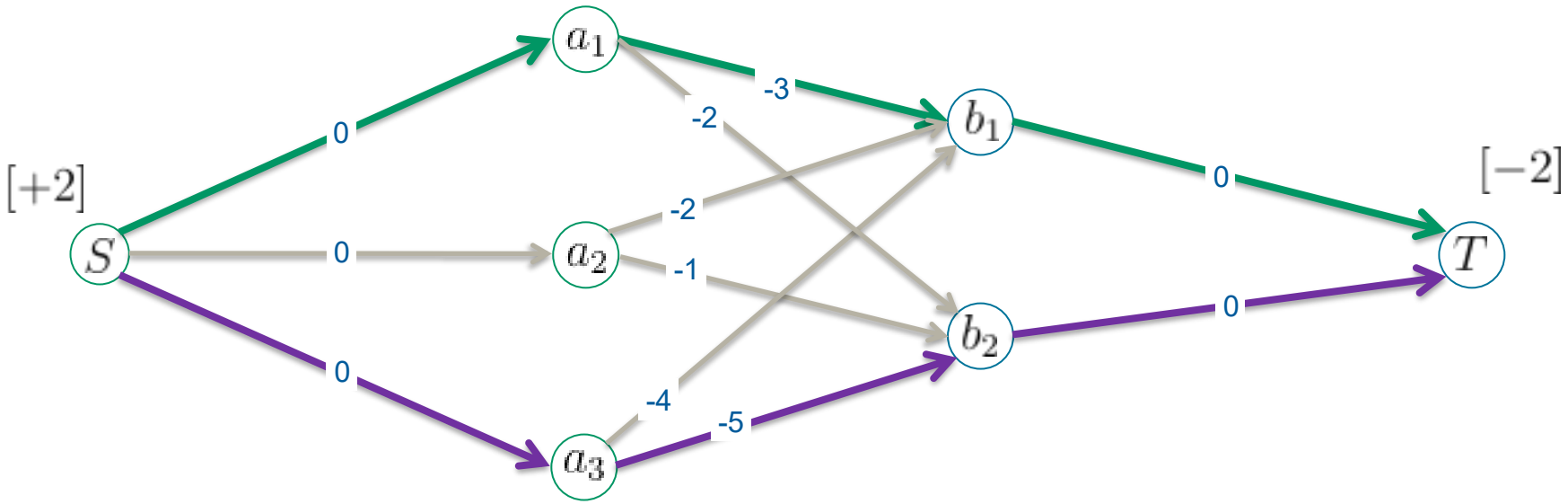
- **Convert into flow graph**
  - Transform weights into costs
  - Add source / target nodes with 0 cost.
  - Directed edges with a capacity of 1.

Minimum-Cost Flow – Small Example



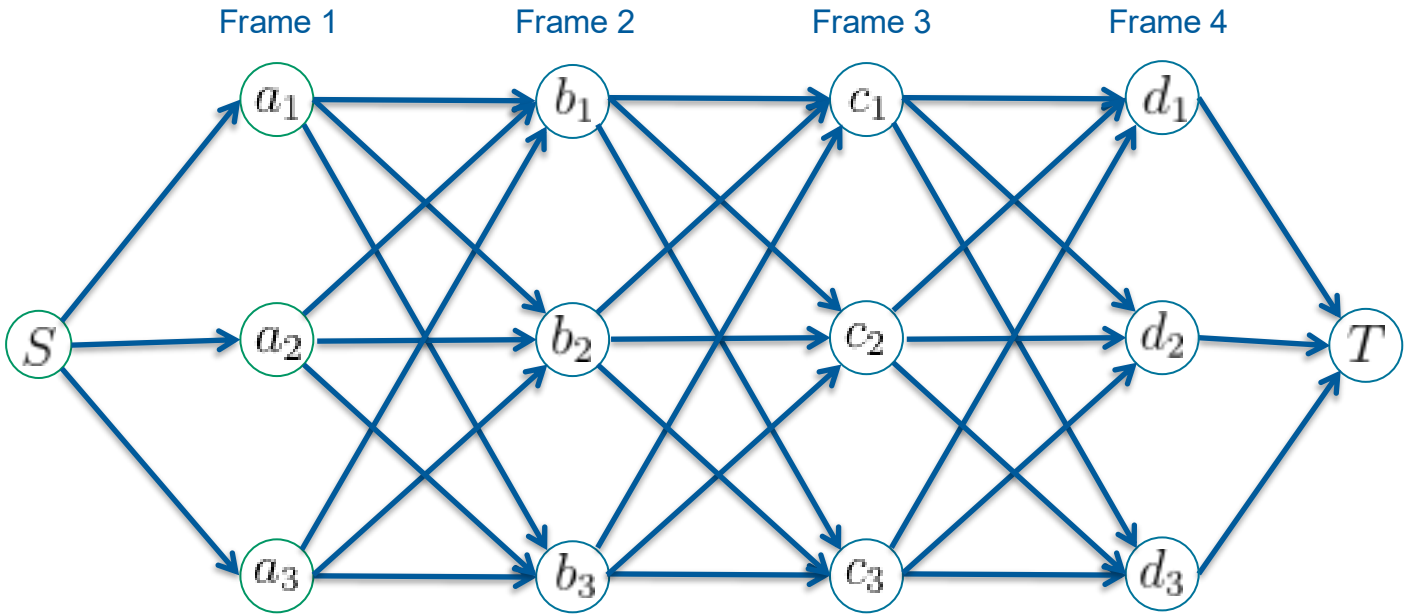
- **Convert into flow graph**
  - Pump N units of flow from source to sink.
  - Internal nodes pass on flow (Σ flow in = Σ flow out).
  - ➔ find the optimal paths along which to ship the flow.

Minimum-Cost Flow – Small Example



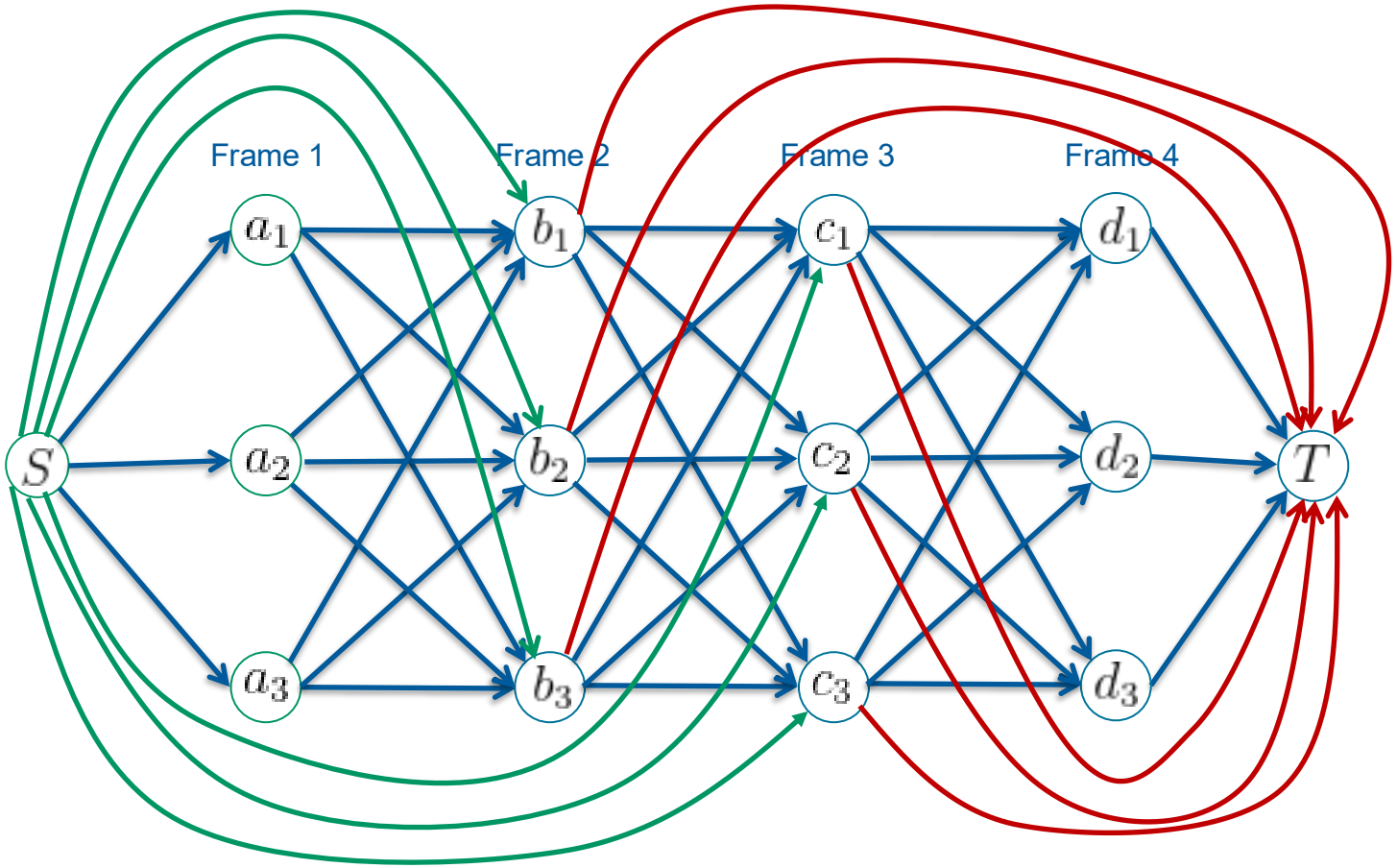
- **Nice property: The min-cost formalism readily generalizes to matching sets with unequal sizes.**

Using Network Flow For Tracking



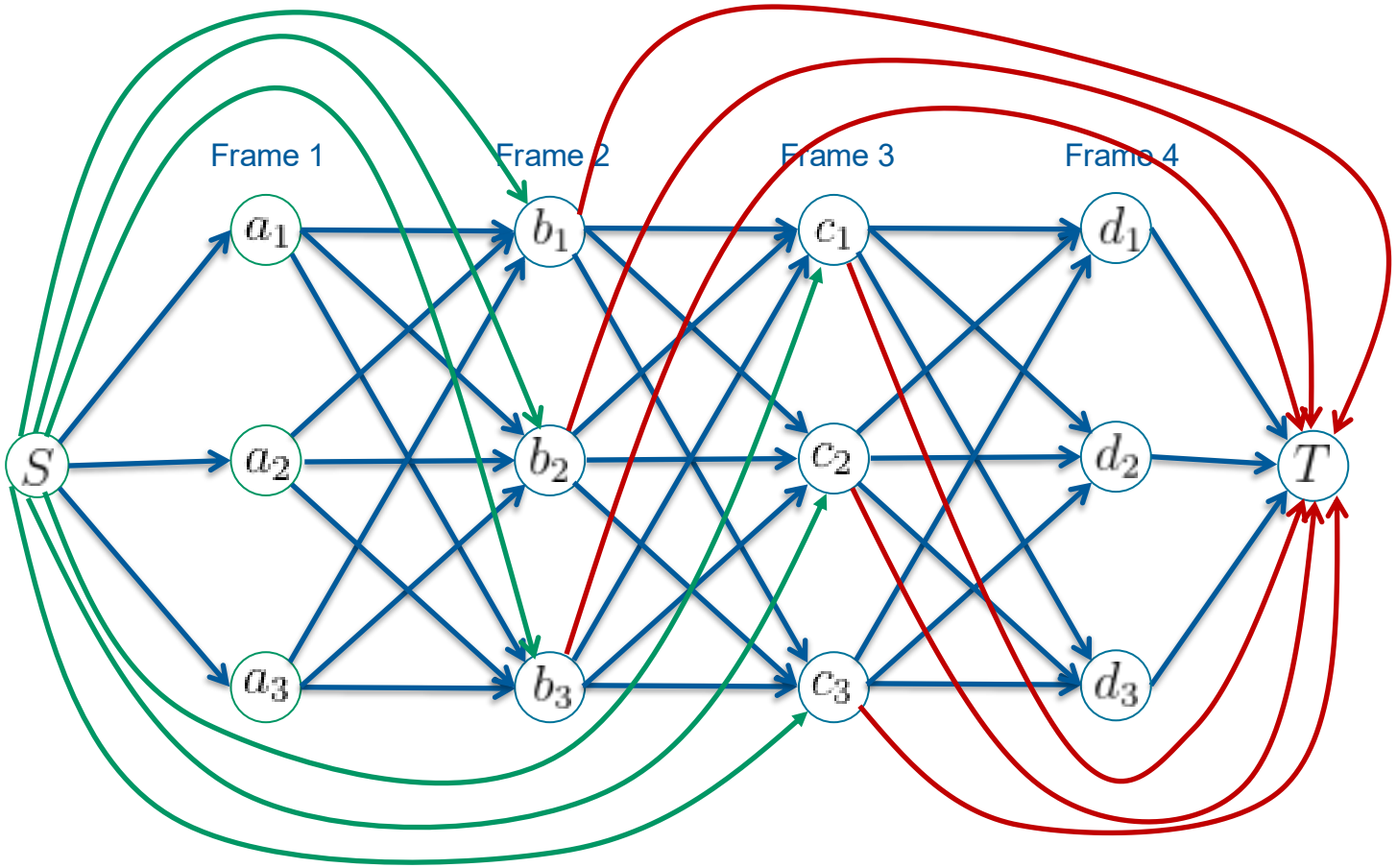
- **Approach:**
  - Seek a globally optimal solution by considering observations over all frames in "batch mode".
  - Extend 2-frame min-cost formulation by adding observations from all frames into the network.

Using Network Flow For Tracking



- Since tracks can start later than Frame 1, and end earlier than Frame 4, connect the source and sink nodes to all intermediate nodes.

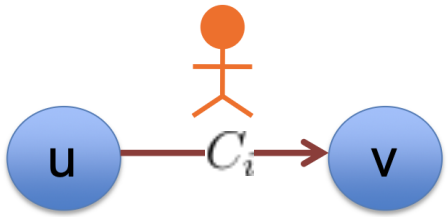
Using Network Flow For Tracking



- Trivial solution: Zero-Cost flow!

Using Network Flow For Tracking [1]

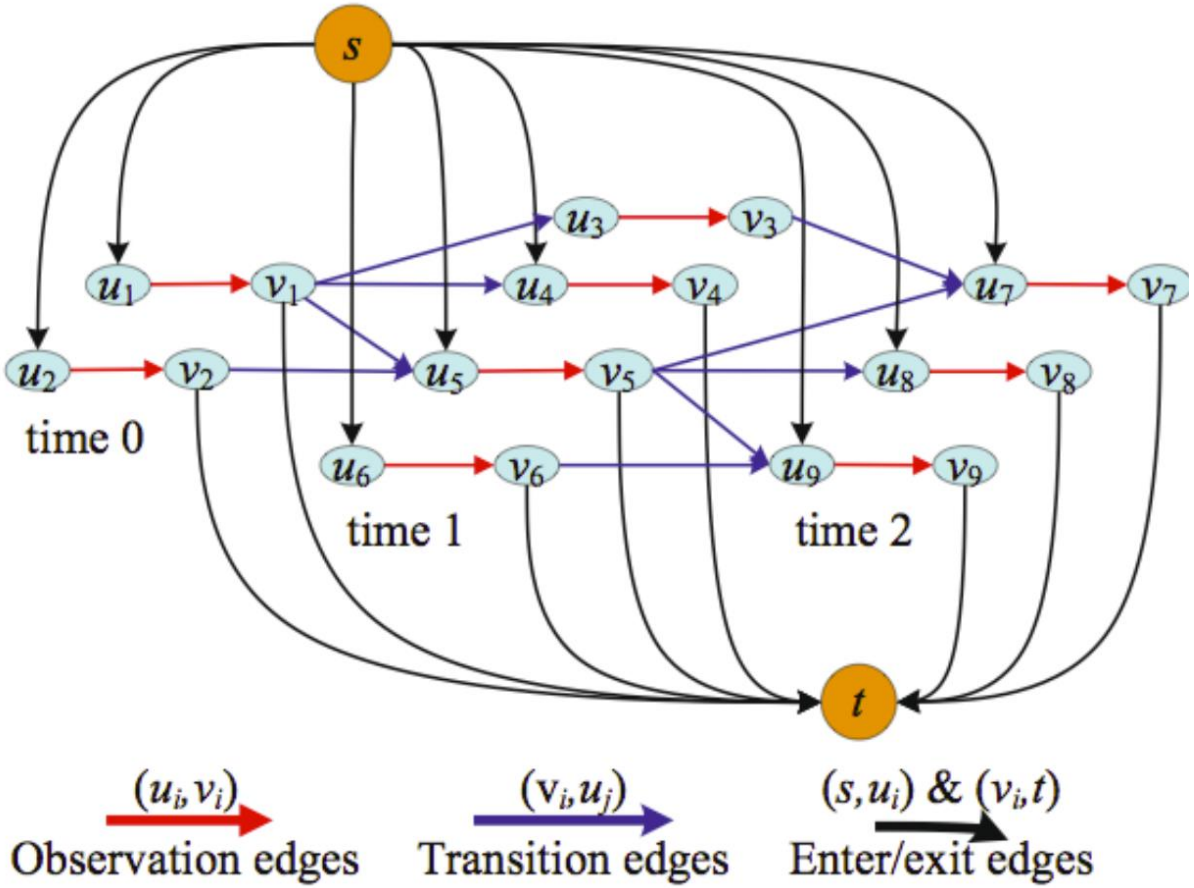
- **Solution: Divide everv detection into 2 nodes**



Detection edge

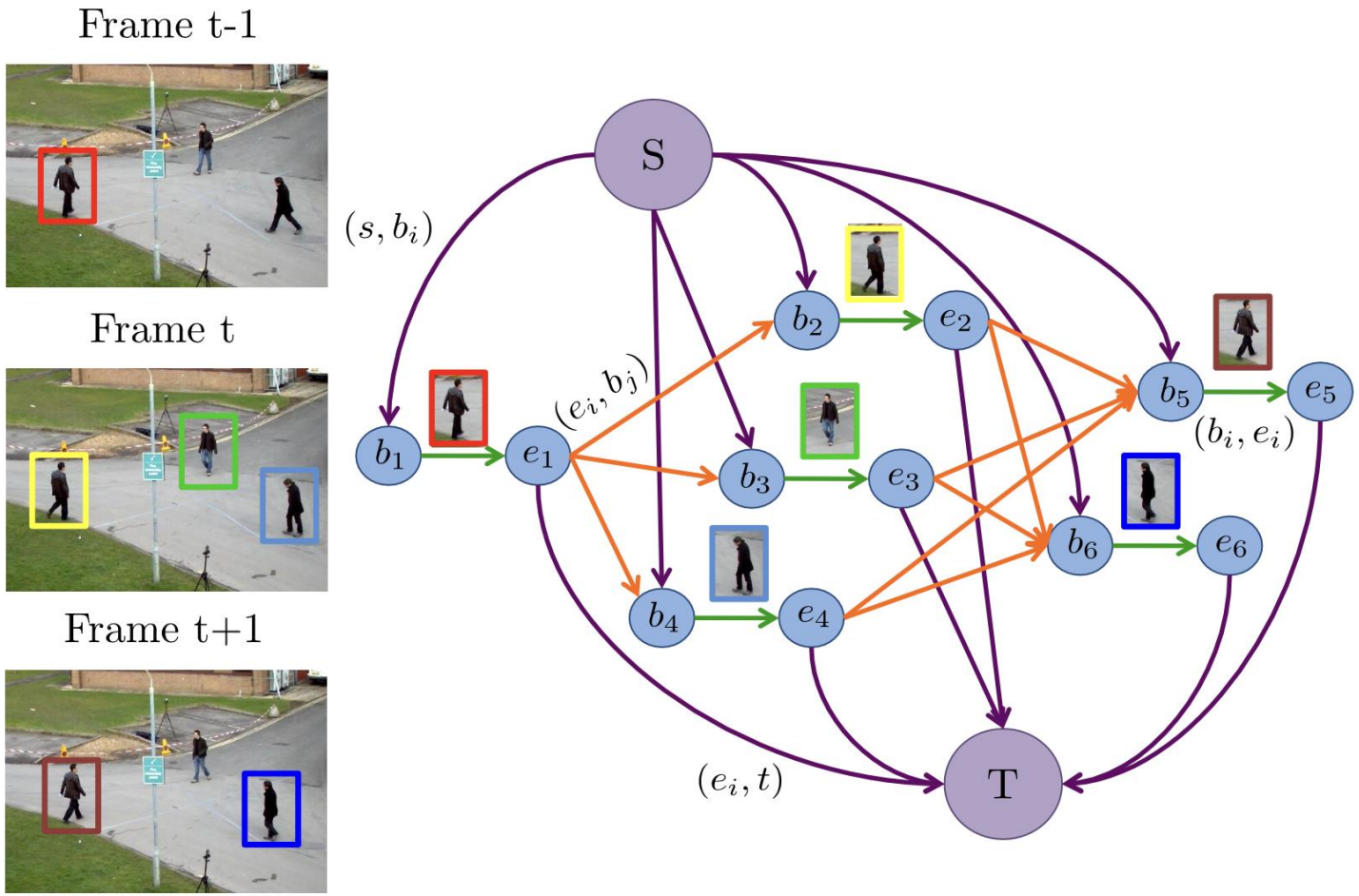
$$C_i = \log \frac{\beta_i}{1-\beta_i} = -\log \frac{1-\beta_i}{\beta_i} = -\log(P_i)$$

with β_i the likelihood that detection i is a false alarm



from [1]

Using Network Flow For Tracking [1]



Minimum Cost Formulation

- **Minimize** $$\mathcal{T}^* = \underset{\mathcal{T}}{\operatorname{argmin}} \sum_i C_{\text{in},i} f_{\text{in},i} + \sum_i C_{i,\text{out}} f_{i,\text{out}} \sum_{i,j} C_{i,j} f_{i,j} + \sum_i C_i f_i$$

subject to

- flow conservation at all nodes: $$\forall i: \; f_{\text{in},i} + \sum_j f_{j,i} = f_i = f_{\text{out},i} + \sum_j f_{i,j}$$

- Edge capacities: $$f_i \leq 1$$

Minimum Cost Formulation

- **Objective Function** $\mathcal{T}^* = \underset{\mathcal{T}}{\operatorname{argmin}} \sum_i C_{\text{in},i} f_{\text{in},i} + \sum_i C_{i,\text{out}} f_{i,\text{out}} \sum_{i,j} C_{i,j} f_{i,j} + \sum_i C_i f_i$

$$\downarrow \quad C_i = -\log(P_i)$$

is equivalent to the Maximum A-Posteriori formulation

$$\mathcal{T}^* = \underset{\mathcal{T}}{\operatorname{argmax}} \prod_i P(\mathbf{o}_i|\mathcal{T}) P(\mathcal{T})$$

Minimum Cost Formulation

- **Objective Function**

$$\mathcal{T}^* = \underset{\mathcal{T}}{\operatorname{argmin}} \underbrace{\sum_i C_{\text{in},i} f_{\text{in},i}}_{\text{In}} + \underbrace{\sum_i C_{i,\text{out}} f_{i,\text{out}}}_{\text{Out}} + \underbrace{\sum_{i,j} C_{i,j} f_{i,j}}_{\text{Transition}} + \underbrace{\sum_i C_i f_i}_{\text{Detection Likelihood}}$$

is equivalent to the Maximum A-Posteriori formulation

$$\mathcal{T}^* = \underset{\mathcal{T}}{\operatorname{argmax}} \prod_i P(\mathbf{o}_i|\mathcal{T}) P(\mathcal{T})$$

Independence Assumption
+ Markov Assumption

$$P(\mathcal{T}) = \prod_{T_t \in \mathcal{T}} P(T_t)$$

Tracking as Network Flow Optimization: Summary

- **Pros**
  - Clear algorithmic framework
  - Equivalence to probabilistic formulation
  - Well-understood Linear Programming optimization problem, efficient algorithms available
  - Globally optimal solution
- **Cons / Limitations**
  - Only applicable to restricted problem setting due to Linear Programming formulation
    - Not possible to encode exclusion constraints between detections (e.g. to penalize physical overlap)
    - Motion model can only draw upon information from pairs of detections (i.e. no constant velocity models)
  - C_{in} and C_{out} fiddly to set in practice (too low ➔ fragmentations, too high ➔ ID switches)

Literature / Links



1. **Zhang, Li, Nevatia:** ***Global data association for multi-object tracking using network flows.*** **https://ieeexplore.ieee.org/document/4587584**



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Thank you very much!